

SOC IP Governance Automation System

All-in-One Report: UI/UX, Features, Flow, Architecture, Statistics, Pros & Cons

Date: 21 February 2026
Confidential

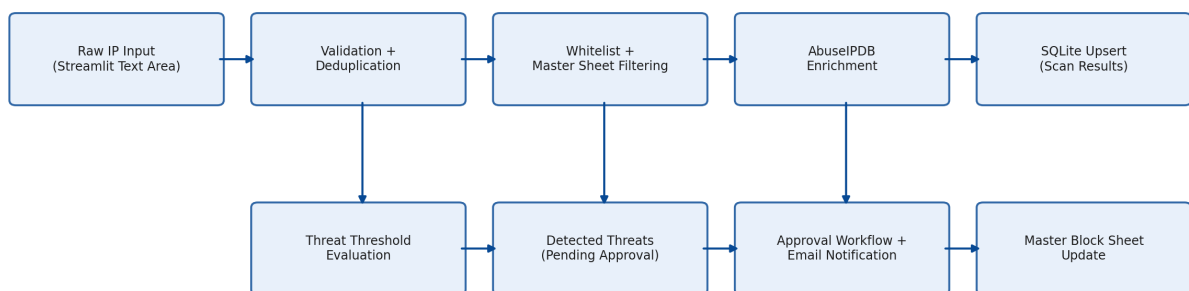
1. Simple Overview (Layman Friendly)

This system helps SOC teams process suspicious IP addresses safely and quickly. Instead of checking everything manually, the platform automatically cleans data, checks threat score, shows approval options, and records final actions in the master block list.

In simple terms: paste IPs → system checks them → analyst approves → records are updated correctly.

2. End-to-End Flow

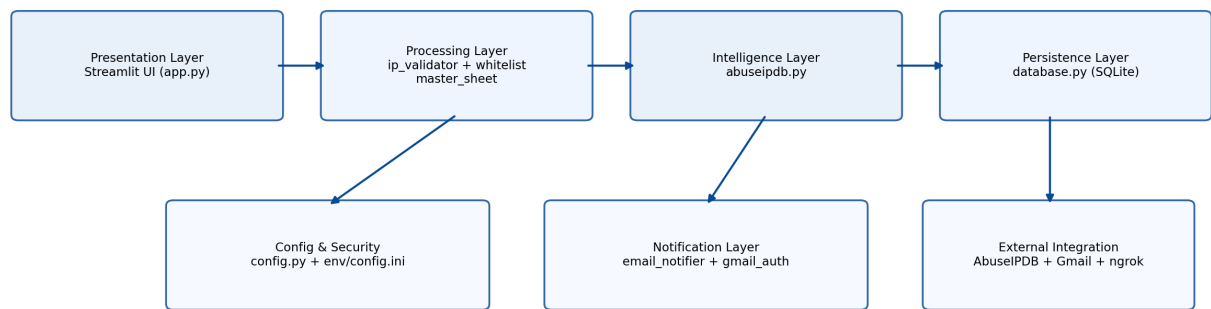
The flow diagram below explains how input data moves through validation, threat checks, and governance updates.



Passive SOC governance pipeline: validate → enrich → approve → document

3. Architecture (How the system is built)

The architecture is modular, so each component has a clear job. This makes the solution stable and easier to maintain.



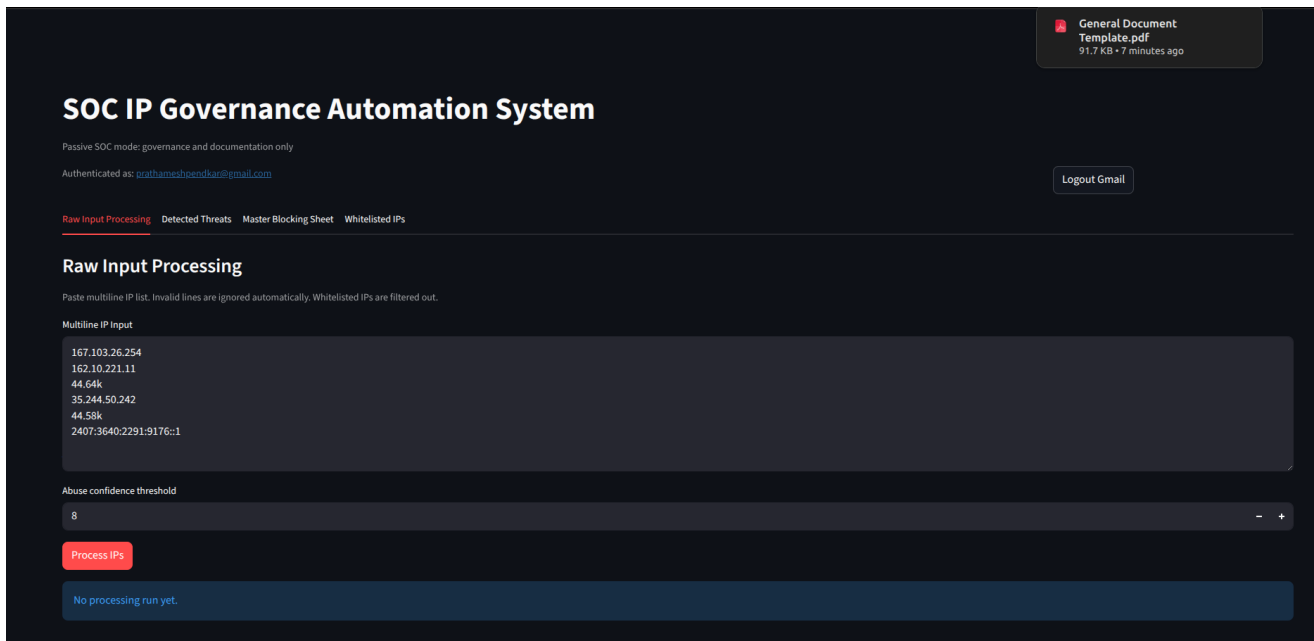
Modular architecture: UI orchestration with isolated integration and persistence responsibilities

4. UI/UX Walkthrough with Screenshots

This section explains each screen in plain language for both technical and non-technical stakeholders.

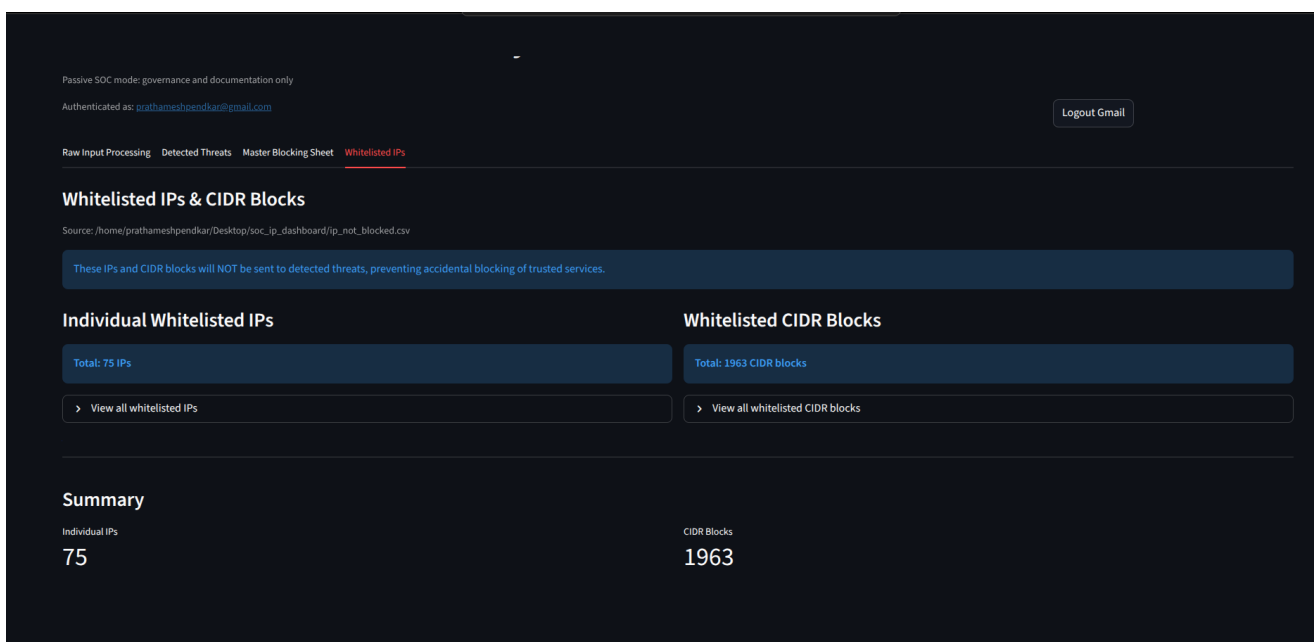
4.1 Login and secure access page

Users first authenticate with Gmail before entering the dashboard. This ensures sender identity and secure access control.



4.2 Raw Input Processing tab

Analyst pastes IP list here. The tool validates data, removes bad lines and duplicates, and prepares clean candidates for checks.



4.3 Detected Threats analysis table

Shows enriched threat data (abuse score, country, ISP, path) so analysts can make informed decisions quickly.

188.166.220.167 (Score: 52)

Approved

Update Status

2602:fb54:1200::16e (Score: 34)

Approved

Update Status

119.13.56.187 (Score: 26)

Approved

Update Status

Bulk Action: Add All Approved IPs to Master Block List

5 IP(s) ready to be added to Master Blocking Sheet and notified via email.

View Approved IPs to be Added

ipAddress	abuseConfidenceScore	countryCode	isp	PATH
134.209.56.230		100 US	DigitalOcean, LLC	/_ 140 /wp-login.php
91.224.92.126		100 LT	Cloud hosting	/_ 140 /wp-login.php
188.166.220.167		52 SG	DigitalOcean, LLC	/_ 140 /wp-login.php
2602:fb54:1200::16e		34 US	Advin Services LLC	
119.13.56.187		26 KH	VIETTEL (CAMBODIA) PTE., LTD	/_ 140 /wp-login.php

Send & Add All to Master Block List

Adds ALL approved IPs to Master Blocking Sheet and sends a single email with all approved IPs in one table.

4.4 Approval Actions and status updates

Each IP can be approved/rejected individually. This keeps decisions transparent and auditable.

Approval Actions

Authenticated sender: prathameshgoendkar@gmail.com | Receiver: prathameshgoendkar2005@gmail.com

Approver Name

Priyanka Maam

Shift

Morning

Default Notes

Approved by SOC

SSO mode: emails are sent via authenticated Gmail account.

Send Test Email

Sends a test approval email using current SMTP settings and selected approver/shift.

Update Individual IP Status

Mark IPs as Approved or Rejected, then use the bulk action button below to add all approved IPs at once.

134.209.56.230 (Score: 100)

Approved

Update Status

91.224.92.126 (Score: 100)

Approved

Update Status

188.166.220.167 (Score: 52)

Approved

Update Status

2602:fb54:1200::16e (Score: 34)

Approved

Update Status

119.13.56.187 (Score: 26)

Approved

Update Status

Bulk Action: Add All Approved IPs to Master Block List

4.5 Bulk approval and add-to-master action

When ready, all approved IPs are added to the master block list in one safe bulk action.

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Edit PATH Column

/__ 140 /wp-login.php

Update PATH

/__ 140 /wp-login.php

Update PATH

/__ 140 /wp-login.php

Update PATH

/__ 140 /wp-login.php

Update PATH

Detected Threats (Above Threshold)

ipAddress	abuseConfidenceScore	countryCode	isp	PATH
134.209.56.230		100 US	DigitalOcean, LLC	/__ 140 /wp-login.php
91.224.92.126		100 LT	Cloud hosting	/__ 140 /wp-login.php
188.166.220.167		52 SG	DigitalOcean, LLC	/__ 140 /wp-login.php
2602:fb54:1200::16e		34 US	Advin Services LLC	
119.13.56.187		26 KH	VIETTEL (CAMBODIA) PTE., LTD	/__ 140 /wp-login.php

Approval Actions

Authenticated sender: [prathameshpendkar@gmail.com](#) | Receiver: [prathameshpendkar2005@gmail.com](#)

Approver Name

Shift

Default Notes

Priyanka Maam

Morning

Approved by SOC

4.6 Master Blocking Sheet view

Displays authoritative blocked records and historical entries for governance and review.

SOC IP Governance Automation System

Passive SOC mode: governance and documentation only

Authenticated as: [prathameshpendkar@gmail.com](#)

Logout Gmail

Raw Input Processing

Detected Threats

Master Blocking Sheet

Whitelisted IPs

Master Blocking Sheet

Source: /home/prathameshpendkar/Desktop/soc_ip_dashboard/IP Blocking - Sheet6.csv

YYYY-MM-DD	Name	IP Address	Reason	HH:MM AM/PM	Approval Status	Notes	Abusive Percentage(Abu	Status(Blocked / Pending)
Date	Shift Member Name	Blocked IP Address	Reason for Blocking	Time of Blocking	Approval Status	Remarks/Additional No	Abusive Percentage(Abu	Status(Blocked / Pending)
YYYY-MM-DD	[Name]	[IP Address]	[E.g., Malicious Activity]	HH:MM AM/PM		[Notes or follow-up acti		
02-01-2026	Harsh Navgale	44.232.229.247	Cloudflare.IP	10.04 PM	Approved	NA	0	Blocked
02-01-2026	Harsh Navgale	35.167.79.216	Cloudflare.IP	10.04 PM	Approved	NA	0	Blocked
02-01-2026	Harsh Navgale	47.128.122.130	Cloudflare.IP	10.04 PM	Approved	NA	87	Blocked
02-01-2026	Harsh Navgale	125.166.117.64	Cloudflare.IP	10.04 PM	Approved	NA	39	Blocked
02-01-2026	Harsh Navgale	202.178.120.157	Cloudflare.IP	10.04 PM	Approved	NA	100	Blocked
02-01-2026	Harsh Navgale	66.92.14.45	Cloudflare.IP	10.04 PM	Approved	NA	0	Blocked
02-01-2026	Harsh Navgale	185.117.225.202	Cloudflare.IP	10.04 PM	Approved	NA	36	Blocked
02-01-2026	Harsh Navgale	35.163.187.192	Cloudflare.IP	10.04 PM	Approved	NA	0	Blocked

4.7 Edit or delete master entries

Allows correction of mistakes and lifecycle management of entries without manual spreadsheet work.

02-01-2026	Puniyat Jangir	103.92.43.96	Cloudflare.IP	4:38:00 PM	Approved	NA	0	Blocked
02-01-2026	Puniyat Jangir	182.64.163.189	Cloudflare.IP	1:30:00 PM	Approved	NA	0	Blocked
02-01-2026	Puniyat Jangir	4.241.216.217	Cloudflare.IP	1:30:00 PM	Approved	NA	100	Blocked
02-01-2026	Puniyat Jangir	20.89.56.154	Cloudflare.IP	1:30:00 PM	Approved	NA	100	Blocked
02-01-2026	Tanmay Kudale	103.131.71.180	Cloudflare.IP	1:43:00 AM	Denied	Monitor further	12	Blocked
02-01-2026	Tanmay Kudale	77.243.90.245	Cloudflare.IP	1:43:00 AM	Approved	NA	0	Blocked
02-01-2026	Tanmay Kudale	31.134.0.11	Cloudflare.IP	1:43:00 AM	Approved	NA	21	Blocked
02-01-2026	Tanmay Kudale	52.10.39.203	Cloudflare.IP	1:43:00 AM	Approved	NA	0	Blocked

Edit or Delete Entries

Select IP to Edit/Delete

Blocked IP Address

Editing IP: Blocked IP Address

Approver Name

Shift Member Name

Notes

Remarks/Additional Notes

Reason

Reason for Blocking

☒ Update Entry

☐ Delete Entry

4.8 Authenticated dashboard home

Once logged in, users access the complete SOC workflow with clear tabs and guided steps.

SOC IP Governance Automation System

Passive SOC mode: governance and documentation only

Gmail authentication is required before accessing the dashboard.

Click Authenticate with Gmail. After successful sign-in, the same Gmail ID will be used as sender email.

[Authenticate with Gmail](#)

4.9 Whitelisted IPs and CIDR view

Trusted IPs/ranges are protected from accidental blocking through a dedicated whitelist control panel.

SOC IP Governance Automation System

Passive SOC mode: governance and documentation only

Authenticated as: prathameshphendkar@gmail.com

Logout Gmail

Raw Input Processing **Detected Threats** Master Blocking Sheet Whitelisted IPs

Detected Threats

Minimum Abuse Confidence Score to Display

20



AbuseIPDB Scanned Results (Abuse Score $\geq$ 20)

 Clear All Results

ipAddress	abuseConfidenceScore	countryCode	isp	PATH
134.209.56.230		100 US	DigitalOcean, LLC	/__ 140 /wp-login.php
91.224.92.126		100 LT	Cloud hosting	/__ 140 /wp-login.php
188.166.220.167		52 SG	DigitalOcean, LLC	/__ 140 /wp-login.php
119.13.56.187		26 KH	VIETTEL (CAMBODIA) PTE., LTD	/__ 140 /wp-login.php

Edit PATH Column

/__ 140 /wp-login.php

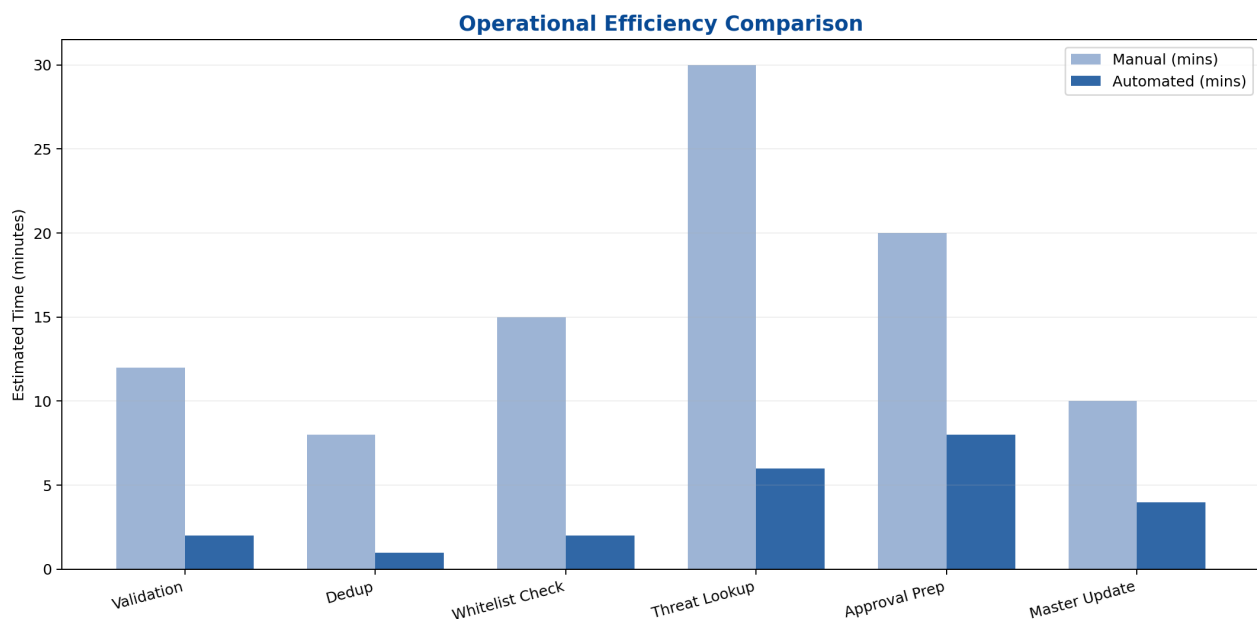
Update PATH

5. Main Features

- Gmail-authenticated access for controlled usage
- IP validation (IPv4 and IPv6), invalid line removal, and deduplication
- Whitelist protection (individual IP and CIDR ranges)
- Already-blocked filtering using master sheet
- AbuseIPDB enrichment (score, country, ISP)
- Threat approval workflow with individual and bulk actions
- Master sheet management with edit/delete support
- SQLite tracking for scan and threat records
- Single-launch operation with Streamlit + ngrok

6. Statistics and Efficiency

The chart below shows why this process is faster than manual handling. Most repetitive tasks are reduced significantly.



What improves	Impact
Validation speed	Automatic checks replace manual line-by-line review
Accuracy	Fewer human errors due to structured workflow
Audit readiness	All key actions are stored in SQLite + master records
Decision quality	Analyst sees score + country + ISP + path before approval
Operational effort	Bulk approval and update actions reduce repetitive clicks

7. Pros and Cons

Pros	Cons
Saves analyst time	Depends on external APIs/services

Simple UI for day-to-day SOC work	Static ngrok endpoint can conflict if active elsewhere
Improved traceability and governance	Still passive mode (not direct firewall automation)
Lower risk of accidental trusted-IP blocking	Spreadsheet-based master source needs disciplined handling

8. Conclusion

This solution makes SOC IP governance faster, safer, and easier to manage. The interface is straightforward for non-technical users, while the backend controls (validation, filtering, enrichment, audit storage) provide technical reliability for SOC teams.